

ME 366 · ELECTROMECHANICAL SYSTEM DESIGN

Project Report

Design, Instrumentation and CFD Validation of a Miniature Open-Circuit Subsonic Wind Tunnel for Low-Reynolds-Number Aerodynamics

GROUP A13

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Abstract

Low-cost wind tunnels can give student teams access to low-Reynolds-number aerodynamic testing, but their measurements are only useful once their accuracy is known. This work describes the design, construction and validation of a miniature open-circuit, induced-draft subsonic wind tunnel with a straw honeycomb, a 6.25:1 fifth-order polynomial contraction and a $160 \times 160 \times 340$ mm test section running at 16–18 m/s ($Re \approx 1.2 \times 10^5$). The tunnel is instrumented with a dual-lever load-cell balance for lift and drag, surface-pressure taps read by a differential pressure sensor, a Pitot-static probe and a glycerine-smoke rake, all logged through an Arduino microcontroller. A NACA 4418 airfoil (-5° to 10°) and a modified Ahmed body were tested and compared with steady $k-\omega$ SST simulations in Ansys Fluent of the confined test section both with and without the model support stand. The tunnel reproduces the computed trends: the zero-lift angle agrees to within 0.1° (-3.7° measured), the measured drag lies within 12% of the with-stand simulations, the Ahmed-body drag coefficient is within 9%, and the pressure distributions match the computed shapes. Without the stand in the simulations, the drag differences grow to 22–80%, showing that the support stand adds significant drag that must be modelled in CFD for small-tunnel validation. The measured lift is 15–20% low at 5° and 10° , and the measured suction is weaker than predicted; these differences are traced to low-Reynolds-number boundary-layer effects, balance pivot coupling and calibration, and pressure-line attenuation, and improvements are proposed for each.

Keywords: subsonic wind tunnel; low Reynolds number; CFD validation; force balance; surface pressure; NACA 4418; Ahmed body

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1 Introduction

1.1 Background and Motivation

Aerodynamic data at Reynolds numbers in the range 10^4 – 10^5 underpin the design of small unmanned aircraft, scale models of ground vehicles and many introductory aerodynamic experiments [1]. In this range the boundary layer stays laminar over much of the surface, and laminar separation, transition and reattachment make the forces sensitive to small changes in Reynolds number, surface finish and freestream turbulence [2]. Such flows are hard to predict, so computations need experimental checks. Wind tunnels that can provide these checks are rarely available to undergraduate teams: commercial teaching tunnels are expensive, and even a recently published open, low-cost design costs several thousand US dollars in hardware [3].

A small tunnel built from commonly available materials can fill this gap, but its measurements are useful only once their accuracy is known, and validating a small facility is difficult. The model blocks a significant fraction of the test section, the support stand is comparable in size to the model and adds drag and interference of its own, and the loads are below one newton, close to the resolution of low-cost sensors. This project addresses both needs. It designs, builds and instruments a miniature open-circuit wind tunnel, and then assesses the tunnel's measurement capability against a computational model of the same confined test section, stand included.

1.2 Related Work

Design guidance for small low-speed tunnels is well established. Mehta and Bradshaw [4] give rules for the settling chamber, honeycomb (cell length of six to eight diameters), screens, contraction ratio and diffuser, and Barlow et al. [5] treat tunnel design, instrumentation and wall corrections in detail. For the contraction, Bell and Mehta [6] showed that a fifth-order polynomial wall, with zero slope and curvature at both ends, balances exit-flow uniformity against the risk of boundary-layer separation near the inlet and exit; the profile is widely used in small tunnels and is adopted here. Ismail et al. [3] published an open design for a low-cost, open-loop suction tunnel for aerodynamic testing at up to 15 m/s, with step-by-step build files. The present tunnel uses the same suction layout at a smaller scale, built largely from laser-cut plywood, 3D-printed parts and hobby-grade electronics, and adds a force balance and surface-pressure measurement.

Two standard test models were chosen because their behaviour is well documented. The NACA four-digit sections are among the most thoroughly catalogued airfoils [7]; the cambered NACA 4418 produces lift at zero incidence, with a zero-lift angle near -4° , which gives a direct check on the lift measurement. The Ahmed body [8] is the standard simplified road vehicle for bluff-body studies; its wake, and hence its drag, is governed by the rear slant angle, the flow over the slant remaining largely attached below a critical angle of about 30° .

For the computations, steady Reynolds-averaged Navier–Stokes (RANS) simulation with the k – ω SST model [9, 10] is the usual engineering choice. The model treats the boundary layer as fully turbulent from the leading edge, however, so it cannot reproduce the laminar separation and transition that dominate at low Reynolds numbers [2]. This limitation matters for the lift comparison in [Section 4.3](#).

1.3 Objectives

The objectives of the project were to:

1. design and build a modular, low-cost, open-circuit subsonic wind tunnel with a 160 mm $\times$ 160 mm test section running at 16–18 m/s, giving $Re \approx 1.2 \times 10^5$ on a 100 mm chord;
2. instrument the tunnel to measure lift, drag, model surface pressures and freestream speed with a low-cost microcontroller data-acquisition system;
3. build a CFD digital twin of the confined test section, with the support stand included, and use it to assess the tunnel's force and pressure measurements on a NACA 4418 airfoil (-5° to 10°) and a modified Ahmed body;

4. identify and, where possible, quantify the sources of disagreement, as a basis for improving the facility; and
5. demonstrate qualitative smoke flow visualization.

1.4 Facility Overview and Key Features

The tunnel (Fig. 1.1) is an open-circuit, induced-draft facility about 1.6 m long, excluding the fan. Room air enters through a straw honeycomb in a 400 mm × 400 mm settling chamber, is accelerated through a 6.25:1 contraction into the 160 mm × 160 mm, 340 mm long test section, and is slowed in a diffuser before being drawn out by a 350 W axial drum fan. Its main features are:

- **Modular, low-cost construction:** flanged plywood and 3D-printed sections screwed together, so the tunnel can be taken apart for transport and any section replaced;
- **Flow conditioning:** drinking-straw honeycomb ($L/D \approx 7$), three screens and a fifth-order polynomial contraction;
- **Three measurement modes:** a dual-lever load-cell balance for lift and drag, surface-pressure taps read by a differential pressure sensor, and a Pitot-static probe for the freestream speed;
- **Interchangeable models:** 3D-printed NACA 4418 and Ahmed-body models with built-in pressure taps, mounted on a stand linked to the balance;
- **Optical access and flow visualization:** an acrylic side window and a removable glycerine-smoke rake;
- **Open tools:** Arduino firmware and MATLAB scripts, with code and CAD shared in a public repository (see [Data and Code Availability](#)); and
- **CFD digital twin:** an Ansys Fluent model of the confined test section, model and stand, used to validate the measurements.

The design, instrumentation and procedures are described in [Section 2](#), the results in [Section 3](#) and their interpretation in [Section 4](#).

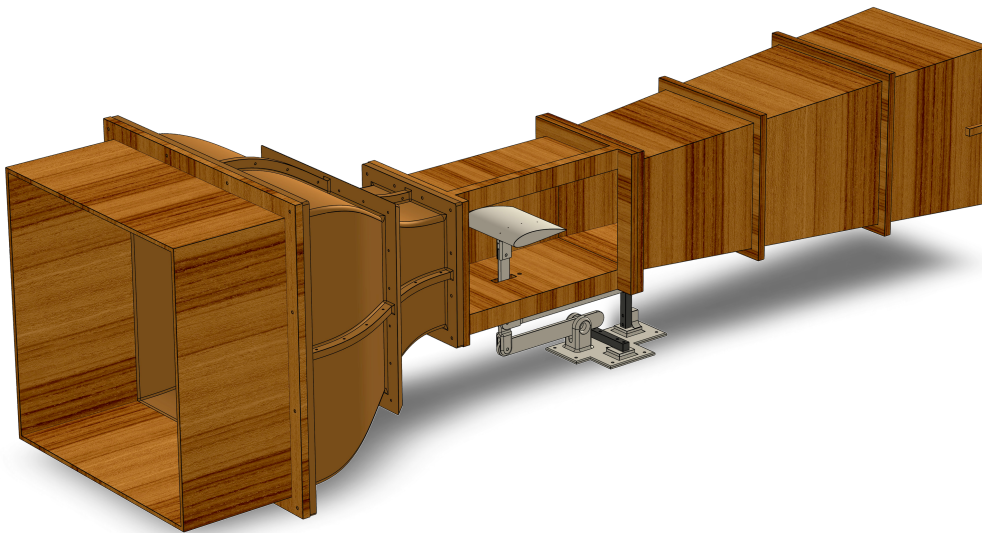


Figure 1.1: CAD model of the wind tunnel with the NACA 4418 and the force balance installed, viewed from the inlet.

1.5 The Project as a Complex Engineering Problem

Under the graduate-attribute framework of the International Engineering Alliance [11], adopted by the Board of Accreditation for Engineering and Technical Education (BAETE), a *complex engineering problem* (CEP) is one that cannot be resolved without in-depth engineering knowledge (attribute P1) and that also

has some or all of attributes P2–P7. This project has all seven attributes, as summarized in [Table 1.1](#). Resolving it also involved complex engineering activities: it drew on diverse resources, from laser cutting, 3D printing and electronics to commercial CFD software (A1); it required resolving interactions between conflicting technical issues, such as the stand’s interference with the drag measurement (A2); it used engineering principles in new ways, including a drinking-straw honeycomb, printed internal tap channels and a balance whose model stand is itself the drag lever (A3); and it went beyond the team’s previous experience through a principles-based approach (A5).

Table 1.1: Attributes of a complex engineering problem [11] and how this project meets them.

Attribute	How the project meets it
P1 Depth of knowledge	Requires engineering fundamentals (K3: fluid mechanics of contractions, diffusers, Pitot measurement and boundary layers; lever statics and load-cell calibration), specialist knowledge (K4: low-Reynolds-number aerodynamics, RANS turbulence modelling and near-wall meshing), design knowledge (K5: honeycomb, screen, contraction and diffuser sizing), knowledge of practice (K6: laser cutting, 3D printing, sensors and microcontroller data acquisition) and the research literature (K8) (Sections 1.2 and 2).
P2 Conflicting requirements	Flow quality against size and cost (contraction ratio and length, number of screens); Reynolds number against blockage in a 160 mm × 160 mm section; balance sensitivity (lever gain) against stiffness; a stand rigid enough to hold the model but small enough to add little drag; one pressure sensor shared between the Pitot probe and the taps; and a fan with enough pressure rise to overcome the conditioning losses at low cost (Sections 2.1, 2.3 and 4.6).
P3 Depth of analysis	No standard solution exists. The contraction wall profile, the dual-lever force resolution and calibration, the CFD model of the confined test section with the stand, the averaging of oscillating solutions and the uncertainty analysis all had to be formulated for this facility, and separating the stand’s drag from the model’s needed a dedicated with/without-stand comparison (Sections 2.5, 2.6 and 4.2).
P4 Infrequent issues	Laminar separation and transition at $Re \approx 10^5$, sub-newton loads near the resolution of low-cost sensors, a support stand comparable in size to the model, steady RANS solutions that settle into limit cycles, and attenuation in long pressure lines (Sections 4.3 to 4.5).
P5 Beyond codes	No code of practice covers the design or validation of a miniature low-cost wind tunnel. The published guidance consists of empirical design rules [4–6], so the validation method, which compares measurements with a CFD model of the same test section, stand included, was developed within the project (Section 2.6).
P6 Diverse stakeholders	Future student users need a facility that is safe, simple to operate and accurate enough to trust; the course sets the schedule and learning outcomes; the budget and the local market limited the materials and the fan that could be used; and the open repository serves others who may reproduce or extend the tunnel (see Data and Code Availability).
P7 Interdependence	The tunnel, force balance, pressure system, data acquisition, test models and CFD model are interdependent sub-problems: the fan choice sets the test speed and Reynolds number, the stand design affects both the balance and the measured drag, the screen position constrains the Pitot probe, and the CFD model must reproduce the stand, the walls and the test conditions for the comparison to be valid (Sections 2 and 4.1).

2 Methodology

2.1 Tunnel Design and Fabrication

The tunnel is an open-circuit, induced-draft (suction) design: a fan at the downstream end draws room air through the tunnel, so the flow reaches the test section before it has passed through the fan. The flow path, shown in Fig. 2.1, is settling chamber with honeycomb, contraction, test section, diffuser, flexible coupling and fan. The plywood panels (6 mm) and the acrylic window (6 mm) were laser-cut from 2D drawings exported from the CAD model, and the panels were bonded with cyanoacrylate adhesive. The contraction, models, balance levers and smoke rake were 3D-printed in PLA on a Bambu Lab A1 printer. The seams were filleted and the surfaces coated with epoxy, and the sections were joined to each other with M3 and M4 screws.

Settling chamber and flow conditioning. The settling chamber has a 400 mm × 400 mm cross-section. It holds a honeycomb made of drinking straws (bubble-tea straws, cell diameter about 10 mm) cut to 70 mm with a hot wire, giving $L/D \approx 7$, within the recommended range of 6–8 [4]; several thousand straws fill the cross-section [TODO: exact straw count]. Three wire-mesh screens are fitted: one at the downstream end of the settling chamber, one at the contraction exit and one downstream of the test section, ahead of the diffuser. [TODO: screen mesh size]

Contraction. The contraction reduces the 400 mm × 400 mm inlet to the 160 mm × 160 mm test section, an area ratio of 6.25:1. Its wall follows the fifth-order polynomial recommended by Bell and Mehta [6], which has zero slope and zero curvature at both ends:

$$y(x) = h_i - (h_i - h_e) \left[10 \left(\frac{x}{L} \right)^3 - 15 \left(\frac{x}{L} \right)^4 + 6 \left(\frac{x}{L} \right)^5 \right], \quad (1)$$

where y is the wall half-height, $h_i = 200$ mm and $h_e = 80$ mm are the inlet and exit half-heights, and $L = 370$ mm is the contraction length ($L/2h_i \approx 0.93$). A doubly curved wall of this size cannot be formed from sheet plywood and is too large for a desktop printer, so the contraction was printed in eight flanged segments and screwed together.

Test section. The test section measures 160 × 160 × 340 mm. It is built from 6 mm plywood with a 6 mm acrylic window for optical access. The model is mounted on a stand passing through the floor to the force balance underneath (Section 2.3). The freestream speed is measured by an ArduPilot-type (APM) airspeed Pitot-static probe [12]. Its 3D-printed mount stands 35 mm downstream of the test-section inlet and 15 mm off the mid-plane towards the window, and holds the probe 44 mm above the floor (Fig. 2.3). The probe points upstream through a hole in the contraction-exit screen, so its tip lies just inside the contraction, clear of the screen wakes, and it is out of the plane of the stand with no part of the model or stand directly downstream of it.

Diffuser and drive. A linear diffuser, 695 mm long, expands the section from 160 mm × 160 mm to 250 mm × 250 mm (area ratio 2.44, wall half-angle 3.7°) and slows the flow before the fan. The diffuser exit is joined to the fan by a flexible coupling, a tarpaulin sleeve tightened with metal cable ties and rope, which keeps fan vibration from reaching the tunnel. The drive is a 14 in, 350 W, 2800 rpm axial drum fan rated at over 2000 CFM, controlled by an AC 220 V, 2000 W SCR voltage regulator. With a model installed it gives 16–18 m/s (57–65 km/h) in the test section; the empty test section reaches about 19 m/s.

2.2 Test Models and Support Stand

Both models were 3D-printed in PLA (Fig. 2.4). Pressure taps are 1 mm holes normal to the surface.

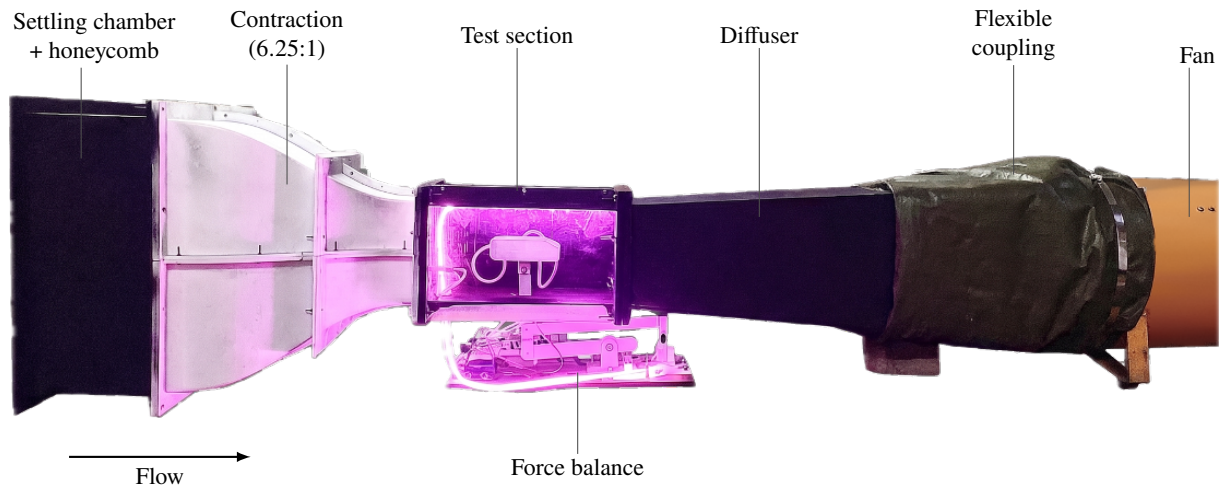


Figure 2.1: The tunnel as built. Air is drawn from left to right by the fan at the downstream end.

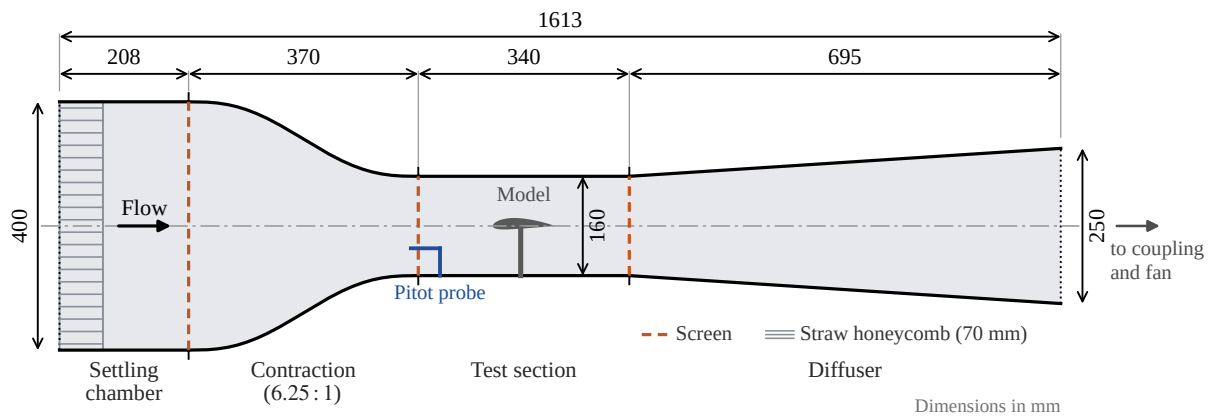


Figure 2.2: Side view of the tunnel with principal dimensions (mm), showing the straw honeycomb, the three screens, the Pitot-static probe and the model position. The flexible coupling and fan downstream of the diffuser are not shown.

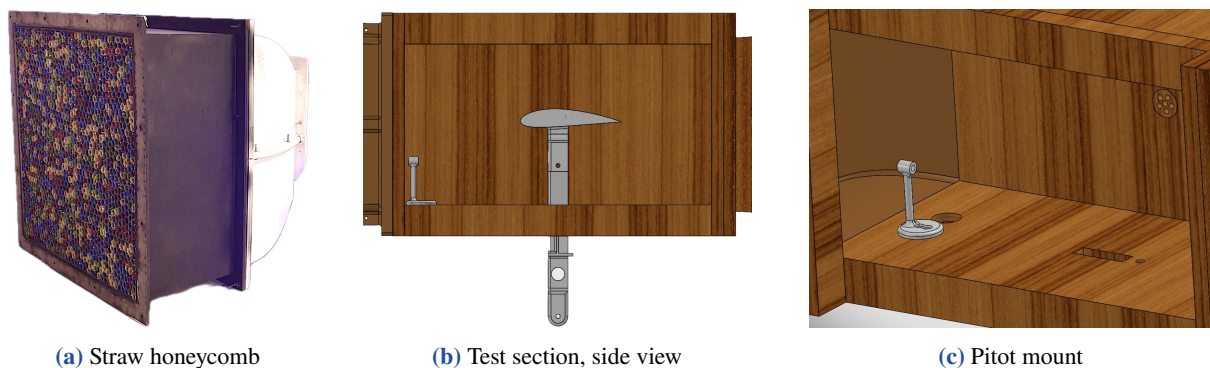


Figure 2.3: Tunnel details. (a) The settling-chamber inlet with the drinking-straw honeycomb. (b) CAD side view of the test section with the NACA 4418 on its stand, the stand's linkage to the balance below the floor, and the Pitot mount near the inlet (left). (c) CAD view of the Pitot mount on the test-section floor; the holes in the far wall are the ports for the pressure-tap tubes.

NACA 4418 airfoil. The airfoil has a chord of $c = 100$ mm and a span of 125 mm, leaving gaps of 10 mm and 25 mm to the window and the opposite wall. The reference area for the force coefficients is the planform area, $A_{\text{ref}} = 0.0125 \text{ m}^2$. Four taps on the upper surface ($x/c = 0.05, 0.20, 0.45, 0.75$) and four on the lower surface ($x/c = 0.12, 0.27, 0.52, 0.68$) lie in one chordwise plane, offset spanwise from the stand; each is connected to an external nozzle by a channel printed inside the model (Fig. 2.4c). The lower tap at $x/c = 0.68$ gave unreliable readings and is excluded, leaving seven taps.

Ahmed body. The bluff-body model is a modified Ahmed body [8], 120 mm long, 40 mm wide and 37 mm high, with a front radius of 11.5 mm. Its rear slant is 28.2 mm long and drops to a 25 mm high base, giving a slant angle of 25° . The body is taller relative to its width than the original, which lets more smoke streaks pass over its upper surface in the visualization runs while keeping a similar frontal area. The reference area is the frontal area, $A_{\text{ref}} = 1.48 \times 10^{-3} \text{ m}^2$. Eight taps run along the upper surface at $x/L = 0.05, 0.13, 0.32, 0.55, 0.72, 0.82, 0.90$ and 0.96 , the last three on the slant (Fig. 2.4d). To avoid the leakage-prone printed internal channels, the Ahmed body was built with its tubes entering from the side and connecting directly behind each tap.

Support stand. Each model sits on a vertical stand whose streamlined sting is 20 mm long in the flow direction, with a 5 mm round leading edge tapering to a 2 mm trailing edge. The sting joins a mount with a $10 \text{ mm} \times 5 \text{ mm}$ through-hole. The NACA model rotates about a pivot about 38 mm below its chord line, so changing the angle of attack also moves it slightly forward or aft. The angle is set by hand with a protractor.

Blockage. At 0° the airfoil's frontal area ($125 \times 18 \text{ mm}$) is 8.8% of the test-section cross-section, or about 10% including the stand, and it grows with angle of attack. The Ahmed body's frontal area is 5.8% of the cross-section. No blockage correction is applied; the CFD reproduces the same confinement (Section 2.6).

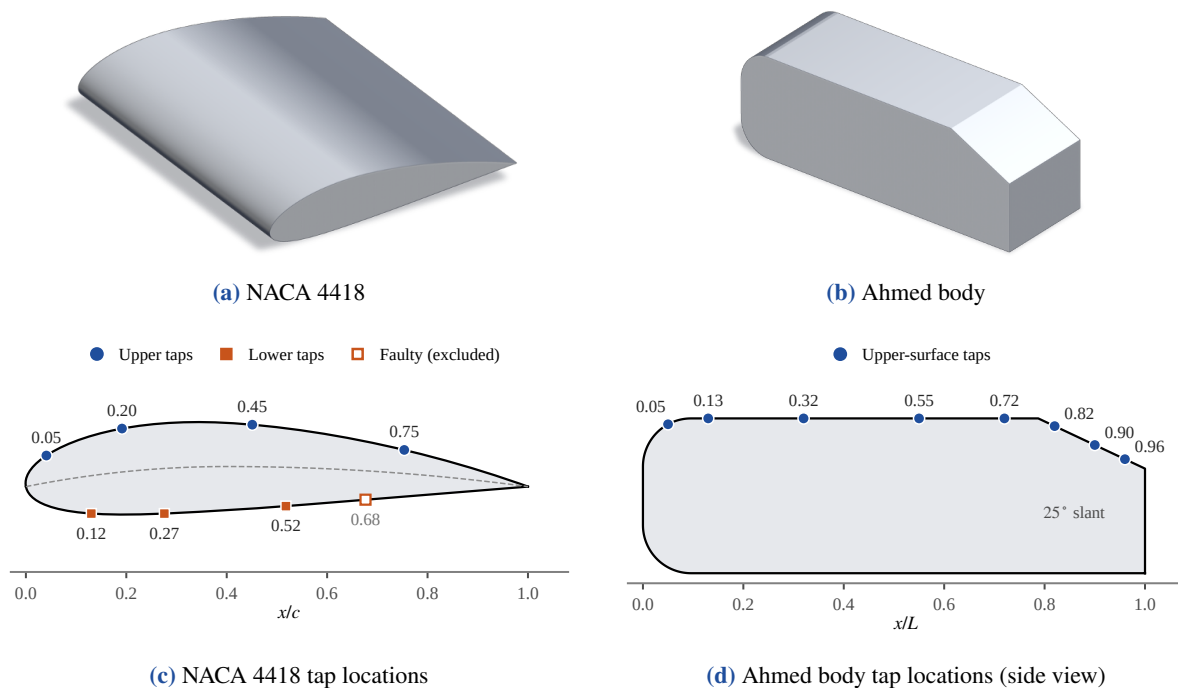


Figure 2.4: Test models (CAD renders) and their pressure-tap positions. In (c) the dashed line is the camber line.

2.3 Instrumentation and Data Acquisition

All sensors are read by an Arduino Mega 2560 [13] and streamed over USB serial to a PC, where the logged data are reduced and plotted in MATLAB Online; key excerpts of the firmware and scripts are given in Section A. The wiring is shown in Fig. 2.6.

Freestream speed. A single MPXV7002DP differential pressure sensor [14] serves both pressure measurements, because the tunnel has one Pitot-static tube, used either to measure the speed or as the static reference for the taps. For speed measurement the Pitot-static tube is connected across the sensor (± 2 kPa, 1 V/kPa at a 5 V supply), read by the Arduino's 10-bit ADC. The sensor output is ratiometric, so its sensitivity is scaled to the measured supply rail (4.93 V). Before every run the zero-pressure output is recorded with the fan off as the mean of 100 samples and subtracted from all later readings. The speed follows from the Pitot pressure difference,

$$U_{\infty} = \sqrt{\frac{2 \Delta p_{\text{Pitot}}}{\rho_{\text{air}}}}, \quad (2)$$

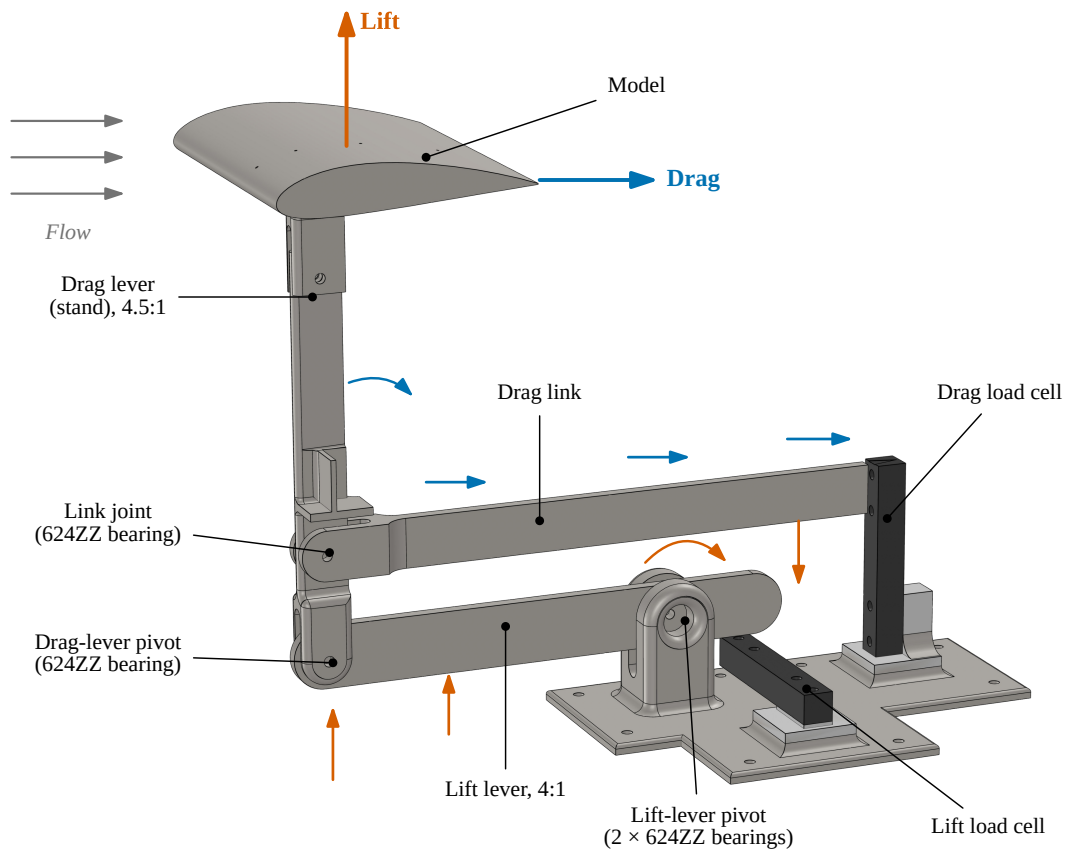
with the density of the tunnel air, $\rho_{\text{air}} = 1.164 \text{ kg/m}^3$ at about 30°C .

Force balance. Lift and drag are measured by a dual-lever balance below the test section (Fig. 2.5). The stand itself is the drag lever: it pivots at its lower end, and a horizontal link attached partway up carries the drag, amplified 4.5:1, to the drag load cell. The stand's pivot sits on one end of the lift lever, which turns about a fixed pivot and presses on the lift load cell with its other end, amplifying the lift 4:1. Every pivot and joint runs on 624ZZ ball bearings: two at the lift-lever pivot and one each at the drag-lever pivot and at the joint between the stand and the drag link. The levers and the link are printed in PLA, and both load cells are 1 kg straight-bar cells, each read by an HX711 24-bit bridge amplifier [15] at 10 samples per second. Each channel was calibrated with the lever system assembled but without the model, so the lever ratio is contained in the calibration slope. The calibration used two points, zero load and one known mass (weighed on a kitchen scale), with the slope stored in the Arduino's EEPROM (Listing A.1). Objects of other known masses were then read correctly to within the $\pm 3 \text{ g}$ (0.03 N) band adopted as the force uncertainty (Section 2.5), and a 31 g reference still read 31 g several weeks later. The balance is re-tared, fan off, at the start of every test. During a run, each output line averages five conversions from each load cell together with the Pitot signal sampled at 1 kHz over the same window (about 500 samples), giving about two readings per second.

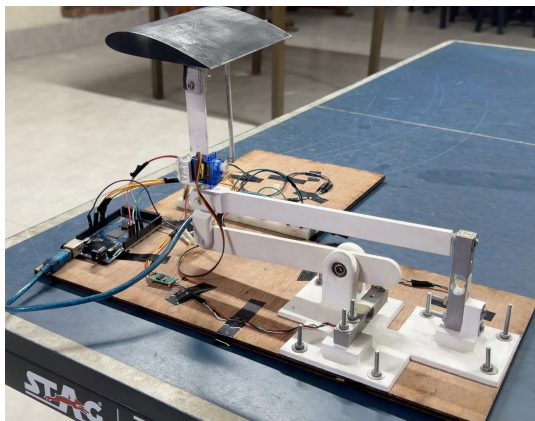
Surface pressure. Each tap is connected by a silicone tube (2.5 mm bore, 5 mm outside diameter, about 0.75 m long) to the positive port of the MPXV7002DP, while the reference port stays connected to the Pitot static port. The sensor therefore reads the tap pressure relative to freestream static pressure directly. The taps are connected one at a time by hand; the firmware streams the tap pressure, each value the mean of 20 samples, and the reading is recorded once it has settled. Because the single sensor is occupied by the tap, the dynamic pressure cannot be measured while the taps are read (Section 2.4).

2.4 Test Procedure and Data Reduction

For each test the model was set to the required angle, the data-acquisition code was started with the fan off to tare the sensors, and the fan was then brought up to speed and allowed to settle before readings were logged. Table 2.1 lists the tests. In the force runs the Pitot tube was connected to the pressure sensor and the speed was recorded together with the forces. In the surface-pressure runs the same sensor was connected between each tap and the Pitot static port, so the dynamic pressure could not be measured at the same time and the nominal test speed was used instead.



(a) CAD model with the load paths



(b) Built, with the NACA 4418



(c) Built, with the Ahmed body and its tubes

Figure 2.5: The dual-lever force balance. (a) Drag turns the stand about its lower pivot and is passed, amplified 4.5:1, through the upper link to the drag load cell; lift turns the lower lever about its pivot and presses, amplified 4:1, on the lift load cell. (b, c) The balance outside the tunnel, with the HX711 amplifiers and the Arduino Mega on the base plate. The servo on the stand belonged to the original angle-of-attack drive and was not used in the final tests.

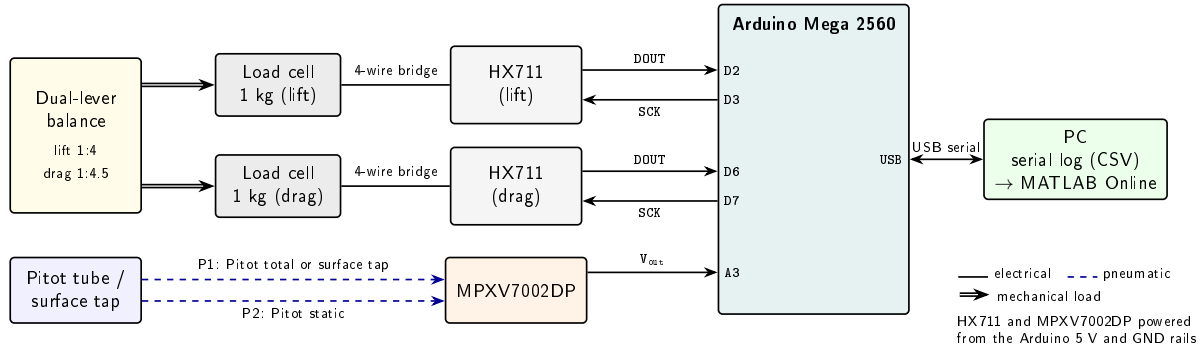


Figure 2.6: Data-acquisition wiring. Both HX711 amplifiers use two digital pins each (data and clock); the pressure sensor feeds analogue input A3. The sensor's reference port stays on the Pitot static port, and its positive port is switched between the Pitot total port and the surface taps.

Table 2.1: Test matrix.

Model	Quantity	α	U_∞ (m/s)
NACA 4418	C_l, C_d	$-5^\circ, 0^\circ, 5^\circ, 10^\circ$	17.5
NACA 4418	C_p (7 taps)	$-5^\circ, 0^\circ, 5^\circ, 10^\circ$	17.5
Ahmed body	C_d	—	16
Ahmed body	C_p (8 taps)	—	16
NACA 4418	Smoke visualization	$-5^\circ, 0^\circ$	—

The forces, in grams, are converted to newtons and reduced to coefficients with the freestream dynamic pressure

$$q_\infty = \frac{1}{2} \rho_{\text{ref}} U_\infty^2, \quad C_l = \frac{L}{q_\infty A_{\text{ref}}}, \quad C_d = \frac{D}{q_\infty A_{\text{ref}}}, \quad C_p = \frac{p - p_\infty}{q_\infty}, \quad (3)$$

where $p - p_\infty$ is the tap reading relative to the Pitot static port. All coefficients, from the tunnel and the CFD alike, use the reference density $\rho_{\text{ref}} = 1.225 \text{ kg/m}^3$; the speeds in Eq. (2) use the actual air density. Because the tunnel air is less dense than ρ_{ref} , this makes q_∞ about 5% larger than the true dynamic pressure at the measured speed; the effect is discussed in Section 4.5. The nominal speeds give $q_\infty = 187.6 \text{ Pa}$ for the NACA 4418 and 156.8 Pa for the Ahmed body. The Reynolds number based on the NACA chord is about 1.2×10^5 , and on the Ahmed body length about 1.3×10^5 (Table 3.1).

2.5 Uncertainty Analysis

The uncertainties are estimates based on resolution and observed reading fluctuation. They do not include sensor non-linearity or drift, tap position errors, angle-setting error, balance cross-coupling or the stand's drag; these systematic effects are discussed in Section 4.5. Table 2.2 lists the inputs. They are propagated by root-sum-square (Listing A.5), using $\delta q_\infty / q_\infty = 2 \delta U / U_\infty$:

$$\frac{\delta C_F}{C_F} = \sqrt{\left(\frac{\delta F}{F}\right)^2 + \left(\frac{2 \delta U}{U_\infty}\right)^2}, \quad \delta C_p = \sqrt{\left(\frac{\delta p}{q_\infty}\right)^2 + \left(C_p \frac{2 \delta U}{U_\infty}\right)^2}, \quad (4)$$

where C_F stands for C_l or C_d and F for the corresponding force. For the lift-to-drag ratio the dynamic pressure cancels, so $\delta(L/D)/(L/D) = \sqrt{(\delta L/L)^2 + (\delta D/D)^2}$.

The resulting uncertainties are ± 0.013 to ± 0.037 in C_l , ± 0.013 to ± 0.015 in the NACA 4418 C_d , ± 0.134 in the Ahmed body C_d , and ± 0.027 to ± 0.136 in C_p . The speed term dominates for the pressure data (a $\pm 14\%$ uncertainty in q_∞), while the force term dominates for small forces such as the drag.

Table 2.2: Uncertainty inputs.

Quantity	Uncertainty	Basis
Force, δF	± 0.03 N (± 3 g)	Observed reading fluctuation; calibration check
Pressure, δp	± 4.9 Pa	One ADC count (4.9 mV) at 1 V/kPa
Speed, force runs	± 0.5 m/s	About twice the Pitot resolution (0.23 m/s at 17.5 m/s)
Speed, pressure runs	± 1.25 m/s	Speed drift of 16–18.5 m/s; q_∞ not measured during tap readings

2.6 Computational Setup (CFD Digital Twin)

The CFD reproduces the tunnel test rather than free flight. The domain is the 160 mm $\times$ 160 mm test section at its real 340 mm length, with the model and its stand in their tunnel positions and no-slip walls on all tunnel surfaces, so blockage and wall effects are the same as in the experiment. The upstream screens and contraction are not modelled; the inlet delivers a uniform flow. Each case was run twice, with and without the stand. The stand-free runs represent the model alone; the with-stand runs are a digital twin of the actual setup and test the measurement system as a whole. The solver and mesh settings are listed in Table 2.3.

For meshing, the CAD was simplified: tap nozzles, tube connectors and the stand's edge fillets were removed, and enclosed dead-air cavities were filled. The sting profile and the 10 mm $\times$ 5 mm through-hole in the mount, through which air flows, were kept exactly. The mesh uses boundary layers (prism cells, added with the *Add Boundary Layers* task in Fluent Meshing) on all model and stand surfaces, with the first-cell height chosen for $y^+ < 1$ (achieved $y^+ \approx 1.7$ overall and 1.9 on the stand), plus offset-surface refinement around the model and stand and local sizing at thin trailing edges and junctions. The student licence's cell limit restricted refinement. The 10° with-stand case used additional wake refinement. No mesh-independence study was carried out.

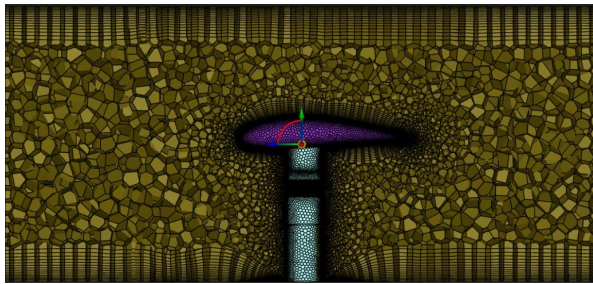
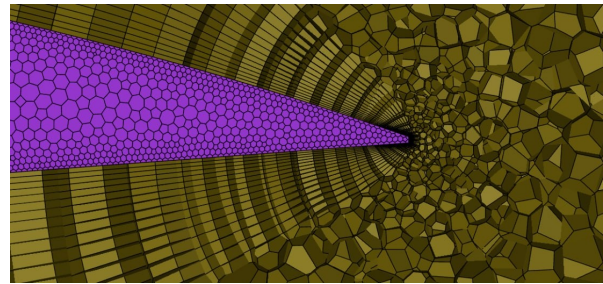
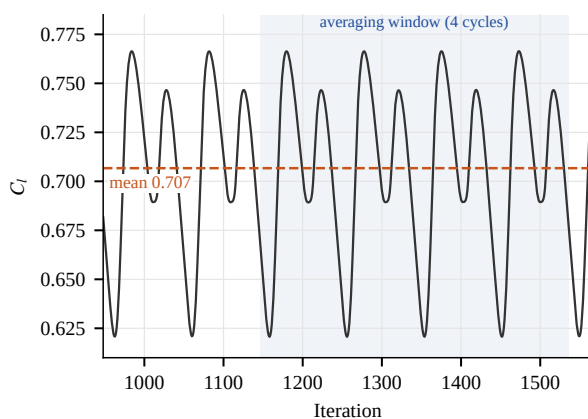
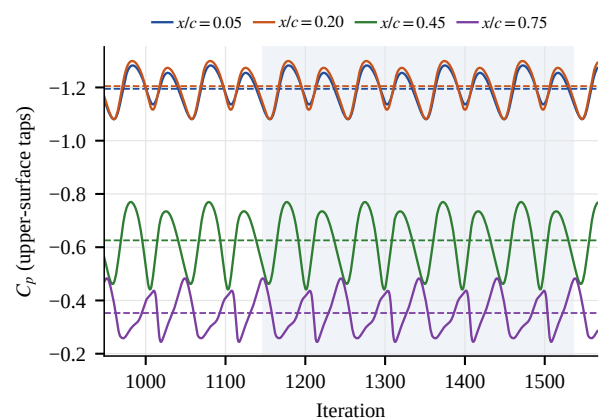
Several solutions, especially those with the stand, did not converge in the residuals because the flow is mildly unsteady. Each run was continued until the C_l , C_d and tap- C_p monitors were constant to three significant figures. Two cases kept oscillating. The stand-free 10° case settled into a regular limit cycle (Fig. 2.8); its cycles were located from the drag-coefficient peaks and the monitors were averaged over the last four whole cycles, accepting the mean only when it changed by less than 0.5% between the last two and the last four cycles (Listing A.3). The Ahmed body's slant pressures oscillated irregularly, so its monitors were averaged over a fixed window after the start-up transient, and the window was checked by comparing the means of its two halves (Listing A.4). In both cases the standard deviation is reported as the CFD spread, and the mean is compared with the experiment because the balance and pressure sensors also time-average. These and all other data-reduction and plotting scripts are available in the project repository (see [Data and Code Availability](#)). Force reports were checked zone by zone to include only model and stand surfaces. Tap C_p values were read from the pressure distribution along the tap plane: the exported curve was rotated into the chord frame, split into upper and lower surfaces, and interpolated at the tap positions (Listing A.2). This replaced single-point probes placed by hand, which gave wrong values at some taps.

2.7 Flow Visualization

Smoke is produced by heating glycerine-soaked cotton with a soldering iron inside a smoke box, which feeds a removable rake of six 3D-printed PLA nozzles (4 mm outlet diameter). The rake sits upstream of the model, offset sideways to clear the stand, and releases streaks that pass over the model. The visualization is qualitative and was not used for any quantitative result.

Table 2.3: CFD solver and mesh settings.

Setting	Value
Software	Ansys Fluent 2026 R1 (Student)
Solver	Pressure-based, steady; Fluent default schemes; energy equation on
Turbulence model	$k-\omega$ SST [9, 10]
Domain	Test section, 160 mm $\times$ 160 mm $\times$ 340 mm, model and stand at tunnel positions
Inlet	Velocity inlet, 17.5 m/s (NACA 4418) or 16 m/s (Ahmed body); default turbulence settings
Outlet	Pressure outlet, default settings (0 Pa gauge)
Walls	No-slip: tunnel walls, model, stand
Fluid	Air, $\rho = 1.225 \text{ kg/m}^3$
Reference values	$\rho = 1.225 \text{ kg/m}^3$; U as inlet; A_{ref} as in Section 2.2
Boundary layers	15–18 layers, growth rate 1.1; $y^+ \approx 1.7$
Volume mesh	Polyhedra (all NACA 4418 cases), poly-hexcore (Ahmed body); growth rate 1.2; minimum orthogonal quality 0.15–0.2 after the <i>Improve Volume Mesh</i> task
Convergence	Monitors constant to 3 s.f., or averaged over whole oscillation cycles

**(a)** Mid-plane cut: test section, model and stand**(b)** Close-up of the trailing edge**Figure 2.7:** Volume mesh for the NACA 4418 at 0° with the stand: polyhedral cells with boundary (prism) layers on the model, the stand and the tunnel walls.**(a)** Lift coefficient**(b)** Upper-surface tap pressure coefficients**Figure 2.8:** Example of a steady solution that settled into a limit cycle instead of converging (NACA 4418, 10° , without stand; period about 98 iterations). The shaded band is the averaging window of four whole cycles and the dashed lines are the resulting means.

2.8 Hardware and Software Components

Table 2.4 lists the main hardware and software components, their ratings and their functions.

Table 2.4: Hardware and software components.

Category	Component	Rating / spec	Function
Structure	Plywood panels	6 mm, laser-cut	Settling chamber, test section, diffuser, frame
Structure	Acrylic window	6 mm	Optical access to the test section
Structure	3D-printed parts	PLA, Bambu Lab A1	Contraction (8 segments), models, levers, Pitot mount, smoke rake
Structure	Fasteners	M3 and M4 screws	Join tunnel sections and balance parts
Structure	Flexible coupling	Tarpaulin sleeve, metal cable ties, rope	Joins diffuser to fan; isolates vibration
Structure	Adhesive and sealant	Cyanoacrylate; epoxy resin	Bond plywood panels; seal seams and coat surfaces
Flow	Straw honeycomb	Several thousand straws [TODO: count] ; 70 mm long, ~10 mm cells, $L/D \approx 7$	Removes swirl and large eddies
Flow	Screens $\times 3$	[TODO: mesh size]	Turbulence reduction (two upstream of the model); one ahead of the diffuser
Flow	Contraction	6.25:1, fifth-order profile	Accelerates and evens the flow
Drive	Axial drum fan	14 in, 350 W, 2800 rpm, 2000+ CFM	Draws air through the tunnel
Drive	SCR voltage regulator	AC 220 V, 2000 W	Sets the fan speed (open loop)
Force	Dual-lever balance	PLA levers; 4:1 lift, 4.5:1 drag	Amplifies sub-newton loads
Force	Ball bearings $\times 4$	624ZZ (4 $\times$ 13 $\times$ 5 mm, shielded)	Lever pivots and link joint (2 at the lift-lever pivot)
Force	Load cells $\times 2$	1 kg, straight bar	Lift and drag measurement
Force	HX711 $\times 2$	24-bit bridge ADC, 10 samples/s	Load-cell signal conditioning
Pressure	MPXV7002DP	± 2 kPa, 1 V/kPa	Pitot pressure and tap pressures
Pressure	Pitot-static probe	APM (ArduPilot) airspeed probe [12], printed mount	Freestream speed; static reference for taps
Pressure	Silicone tubing	2.5 mm bore, 5 mm OD, about 6 m in 8 lengths	Connects taps to the sensor
DAQ	Arduino Mega 2560	ATmega2560, 5 V, 10-bit ADC	Sampling, taring, serial output
Flow vis.	Smoke rake	6 nozzles, 4 mm outlet; glycerine smoke	Qualitative flow visualization
Software	Arduino firmware	C++, Arduino IDE	Data acquisition and taring (see repository)
Software	MATLAB Online	–	Data reduction and plotting (see repository)
Software	Ansys Fluent	2026 R1 Student	CFD digital twin
Software	SolidWorks	2026	Design of the tunnel, balance and models

3 Results

Wind-tunnel (WT) measurements are compared with two CFD solutions of the confined test section: one that includes the physical model stand (“with stand”) and one of the model alone (“without stand”). Unless stated otherwise, percentage differences are $\Delta = (\text{WT} - \text{CFD})/|\text{CFD}| \times 100\%$, and WT uncertainties are the resolution/fluctuation-based estimates of [Section 2.5](#).

3.1 Test Conditions

The NACA 4418 was tested at $U_\infty = 17.5$ m/s and the Ahmed body at 16 m/s; the test conditions are listed in [Table 3.1](#).

Table 3.1: Nominal test conditions ($\rho = 1.225$ kg/m³).

Quantity	NACA 4418	Ahmed body
Reference length	chord $c = 100$ mm	length $L = 120$ mm
Reference area A_{ref}	0.0125 m ²	1.48×10^{-3} m ²
Nominal U_∞	17.5 m/s	16.0 m/s
Dynamic pressure q_∞	187.6 Pa	156.8 Pa
Reynolds number	$\approx 1.2 \times 10^5$ (on c)	$\approx 1.3 \times 10^5$ (on L)
Mach number	0.051	0.047
Angles of attack	$-5^\circ, 0^\circ, 5^\circ, 10^\circ$	0°

3.2 NACA 4418: Aerodynamic Forces

[Table 3.2](#) lists the measured lift and drag coefficients and lift-to-drag ratios together with both CFD solutions; they are plotted in [Figs. 3.1](#) and [3.2](#).

Table 3.2: NACA 4418 force coefficients: wind tunnel (WT) vs. CFD with and without the model stand. $\Delta = (\text{WT} - \text{CFD})/|\text{CFD}|$.

	α	Wind tunnel		CFD (with stand)		CFD (without stand)	
		Value	$\pm$	Value	Δ (%)	Value	Δ (%)
C_l	-5°	-0.0624	0.0130	-0.0717	+13.0	-0.101	+38.0
	0°	0.176	0.0161	0.177	-0.7	0.173	+1.6
	5°	0.354	0.0238	0.417	-15.1	0.416	-14.8
	10°	0.612	0.0372	0.762	-19.7	0.707	-13.4
C_d	-5°	0.0636	0.0131	0.0723	-12.0	0.0390	+63.0
	0°	0.0753	0.0133	0.0734	+2.6	0.0418	+80.2
	5°	0.0912	0.0136	0.0854	+6.8	0.0656	+39.1
	10°	0.155	0.0154	0.142	+9.1	0.128	+21.6
C_l/C_d	-5°	-0.981	0.276	-0.992	+1.1	-2.58	+62.0
	0°	2.33	0.423	2.41	-3.2	4.14	-43.6
	5°	3.88	0.552	4.88	-20.5	6.34	-38.8
	10°	3.95	0.330	5.36	-26.3	5.54	-28.8

Lift. At 0° the measured lift coefficient agrees with the with-stand CFD to within 0.7% ($C_l = 0.176$ vs. 0.177). At -5° the measured value is 13% smaller in magnitude, and at 5° and 10° it falls 15.1% and

19.7% below the with-stand CFD; at these two angles the gap (0.063 and 0.150) is 2.6 and 4.0 times the WT uncertainty. Including the stand changes the predicted lift little at 0° and 5° (under 3%) but raises it by 8% at 10° and reduces its magnitude by 29% at -5° . Linear interpolation between -5° and 0° gives a zero-lift angle of -3.7° for the WT and -3.6° for the with-stand CFD. Between 0° – 5° and 5° – 10° the lift-curve slope increases in both data sets, from 0.036 to 0.052/ $^\circ$ (WT) and from 0.048 to 0.069/ $^\circ$ (CFD with stand).

Drag. The measured drag agrees with the with-stand CFD at every angle within the WT uncertainty, with differences of -12.0% to $+9.1\%$. Against the stand-free CFD the WT drag is 22–80% higher; the absolute excess is nearly constant, $\Delta C_d = 0.025$ – 0.034 (mean 0.028), whereas the stand increment predicted by CFD (with-stand minus without-stand) decreases with angle from 0.033 at -5° to 0.015 at 10° .

Lift-to-drag ratio. At -5° and 0° the measured C_l/C_d matches the with-stand CFD within 3.2% (Fig. 3.2). At higher angles the measured ratio levels off, from 3.88 at 5° to 3.95 at 10° , while the with-stand CFD keeps rising (4.88 to 5.36), leaving the WT 20–26% lower. The stand-free CFD, lacking the stand drag, predicts a much higher ratio that peaks at 5° (6.34).

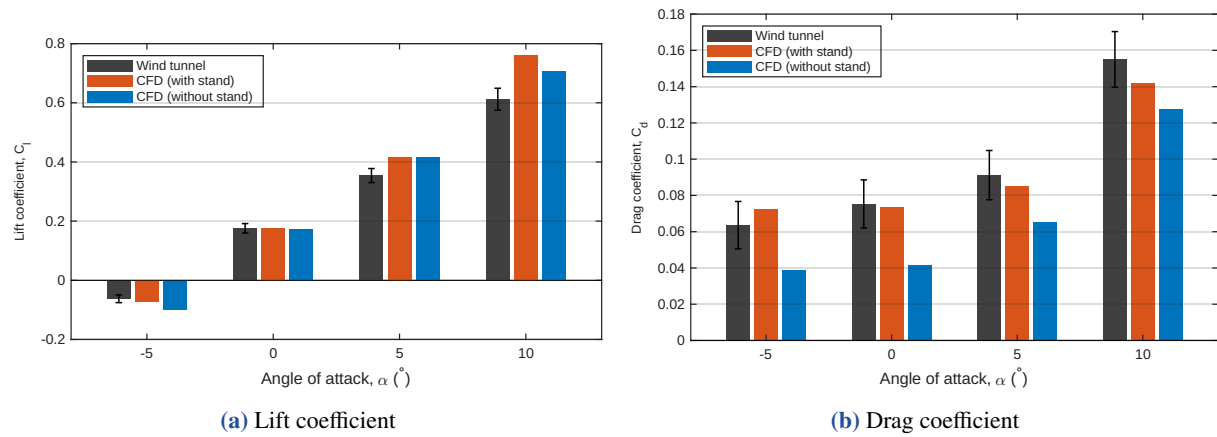


Figure 3.1: NACA 4418 force coefficients vs. angle of attack: wind tunnel (error bars: estimated uncertainty) and CFD with and without the model stand.

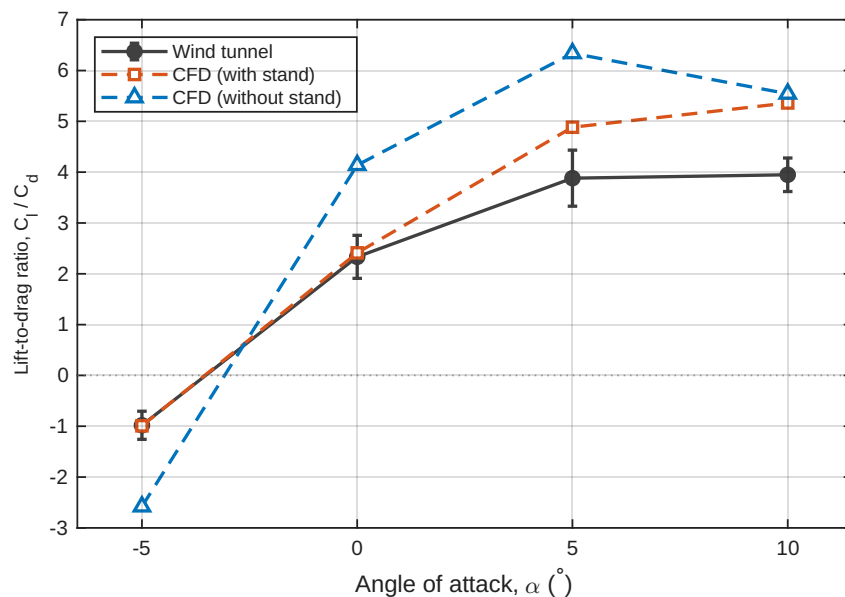


Figure 3.2: NACA 4418 lift-to-drag ratio vs. angle of attack.

3.3 NACA 4418: Surface Pressure Distributions

Figure 3.3 compares the measured tap pressures with the continuous with-stand CFD distribution on the tap plane at each angle; Fig. 3.4 regroups the same tap data by surface to show the effect of angle of attack. The differences at each tap are listed in Table 3.3, and the full measured and computed tap values in Section B. The faulty lower tap at $x/c = 0.68$ is excluded.

The measured distributions reproduce the CFD shapes and their change with angle of attack: upper-surface suction grows and lower-surface suction decreases as α increases, and the suction peak sits at the same tap as in the CFD (the $x/c = 0.20$ upper tap at 0° and 5° , the leading upper tap at 10° , and the leading lower tap at -5°). In magnitude, however, the WT reads less suction (higher C_p) than the CFD at 23 of the 28 taps; the exceptions are the leading upper tap at -5° and 0° and the three lower taps at 10° . Only 3 of the 28 differences lie within the WT uncertainty. The gap is largest where the suction is strongest: 0.19–0.25 at the suction-peak tap for -5° to 5° , rising to 0.45–0.51 at the two leading upper taps at 10° . The rearmost upper tap ($x/c = 0.75$) reads 0.16–0.23 less suction than the CFD at every angle. On the lower surface at 10° the WT shows slight suction ($C_p = -0.05$ to -0.11) where the CFD predicts small positive pressures (+0.01 to +0.17), i.e. the sign of C_p differs.

Table 3.3: NACA 4418 tap pressure difference $C_p^{\text{WT}} - C_p^{\text{CFD}}$ (CFD with stand); WT uncertainty in parentheses. Positive values mean the WT C_p is higher (less suction) than the CFD. Entries within the WT uncertainty are marked *.

Surface	x/c	$\alpha = -5^\circ$	0°	5°	10°
Upper	0.05	-0.056 (0.044)	-0.035 (0.041)*	+0.003 (0.106)*	+0.510 (0.136)
Upper	0.20	+0.176 (0.049)	+0.231 (0.085)	+0.188 (0.117)	+0.451 (0.128)
Upper	0.45	+0.155 (0.061)	+0.216 (0.070)	+0.117 (0.084)	+0.216 (0.083)
Upper	0.75	+0.158 (0.029)	+0.181 (0.028)	+0.159 (0.027)	+0.233 (0.027)
Lower	0.12	+0.249 (0.099)	+0.303 (0.050)	+0.083 (0.030)	-0.219 (0.027)
Lower	0.27	+0.197 (0.057)	+0.219 (0.039)	+0.061 (0.032)	-0.146 (0.031)
Lower	0.52	+0.096 (0.036)	+0.104 (0.032)	+0.015 (0.030)*	-0.114 (0.030)

3.4 Ahmed Body

Drag. The measured drag coefficient is $C_d = 0.671 \pm 0.134$, 9.0% above the with-stand CFD (0.616 ± 0.004 , 1 SD of the averaged monitor) and within the WT uncertainty (Table 3.4). The stand-free CFD predicts $C_d = 0.443$, which the WT exceeds by 51.5%.

Table 3.4: Ahmed body drag coefficient. WT $\pm$: estimated uncertainty; CFD $\pm$: 1 SD of the averaged monitor.

Source	C_d	$\pm$	Δ vs. WT (%)
Wind tunnel	0.671	0.134	–
CFD (with stand)	0.616	0.0035	+9.0
CFD (without stand)	0.443	–	+51.5

Surface pressure. Along the upper surface (Fig. 3.5) the WT reproduces every feature of the CFD distribution: the suction peak behind the nose ($x/L = 0.05$), the recovery to a near-flat roof plateau ($x/L \approx 0.3$ – 0.55), the second suction peak near the roof–slant junction (highest at the $x/L = 0.82$ tap) and the recovery down the slant. The WT suction is weaker than the CFD at every tap, by 0.14–0.42 in C_p : the difference is largest behind the nose (0.42 and 0.37 at $x/L = 0.05$ and 0.13), 0.19–0.29 over the slant, and smallest on the roof (0.14–0.15). The ratio WT/CFD varies between taps from 0.19 to 0.61, so the difference is not a uniform scaling.

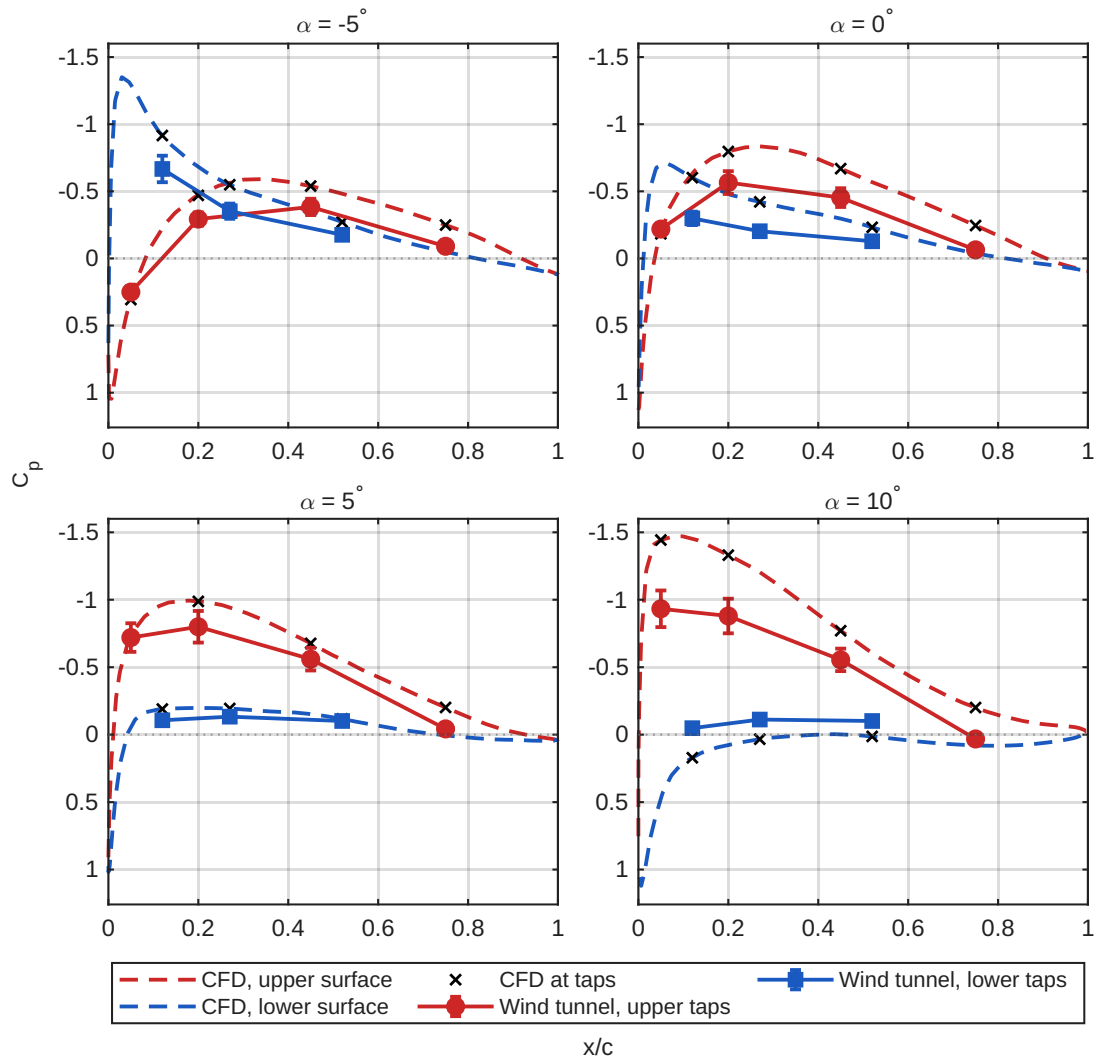


Figure 3.3: NACA 4418 surface pressure coefficient at $\alpha = -5^\circ$, 0° , 5° and 10° : wind-tunnel taps (error bars: estimated uncertainty) vs. the with-stand CFD distribution on the tap plane.

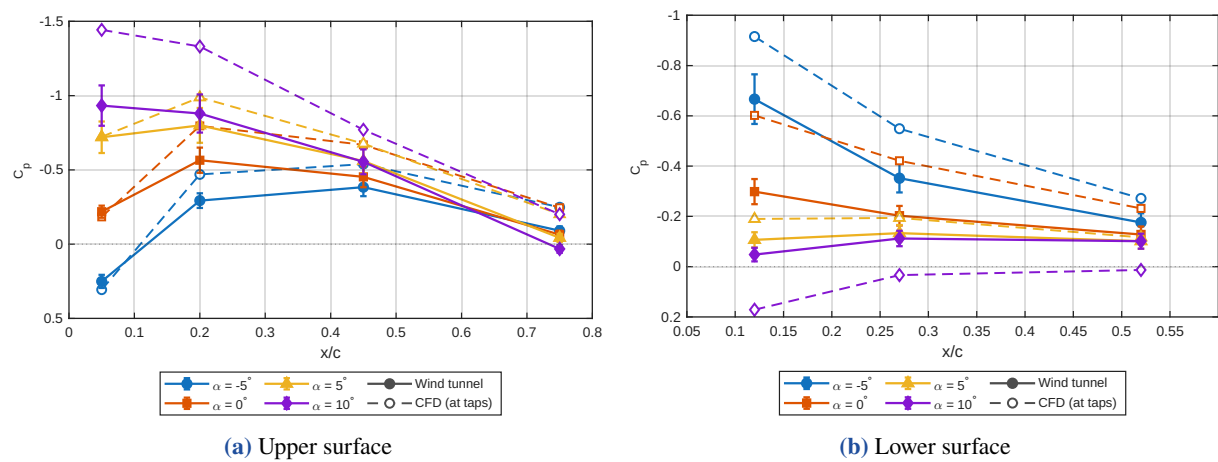


Figure 3.4: NACA 4418 tap pressures grouped by surface: wind tunnel (solid) vs. with-stand CFD at the taps (dashed) for each angle of attack.

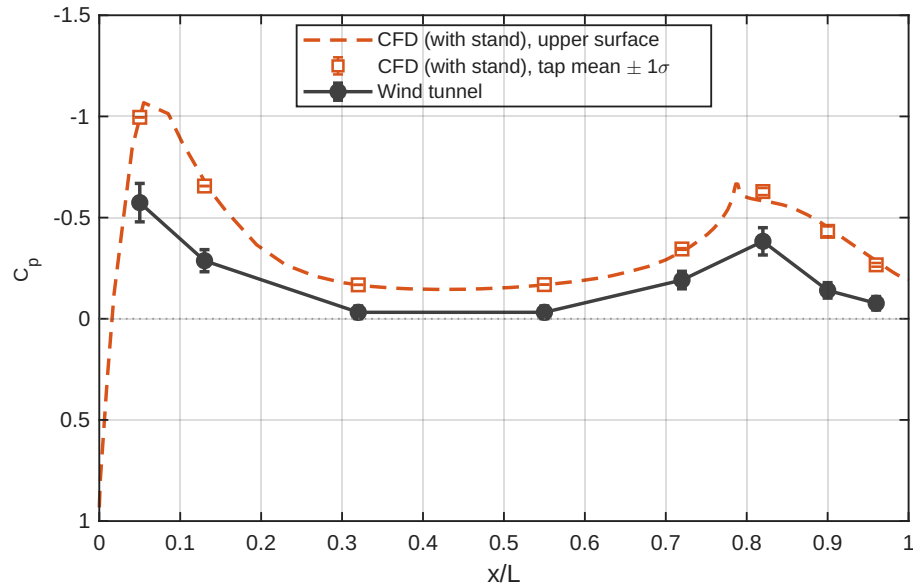
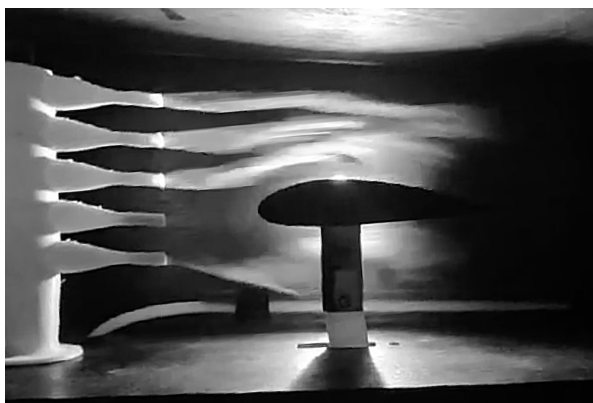


Figure 3.5: Ahmed body upper-surface pressure coefficient along its length: wind tunnel vs. with-stand CFD (continuous curve and tap means $\pm 1\sigma$ of the averaged solution).

3.5 Flow Visualization

Smoke visualization was used as a qualitative demonstration of the facility's flow-visualization capability, complementing rather than replacing the force and pressure measurements. Representative images of the NACA 4418 at $\alpha = 0^\circ$ and -5° are shown in Fig. 3.6; in both, the smoke rake is at the left and the flow is from left to right.

At 0° (Fig. 3.6a) the smoke filaments passing over the model follow both surfaces to the trailing edge with no visible separation, consistent with the smooth pressure recovery measured on both surfaces at this angle (Fig. 3.3). At -5° (Fig. 3.6b) the filaments above the model stay close to and follow the upper surface, while below it the smoke does not follow the lower surface and becomes diffuse, suggesting separated or unsteady flow on the lower side. The lower-surface taps, which extend only to $x/c = 0.52$, still show a steady pressure recovery at this angle, so any such separation would lie further aft.



(a) $\alpha = 0^\circ$



(b) $\alpha = -5^\circ$

Figure 3.6: Glycerin smoke-rake visualization of the NACA 4418 (flow from left to right). Images converted to greyscale, sharpened for contrast and cropped to a common frame; no other processing.

4 Discussion

4.1 Overall Agreement with CFD

The tunnel reproduces every trend predicted by the with-stand CFD, while the magnitudes differ by amounts that can largely be traced to identifiable causes. For the NACA 4418, lift at 0° agrees to within 0.7%, the measured drag lies within -12% to $+9\%$ of the computed values, and both data sets place the zero-lift angle near -3.7° (WT) and -3.6° (CFD), close to the $\approx -4^\circ$ expected for a 4%-camber section [7]. Both lift curves steepen between 5° and 10° , and the measured pressure distributions keep the CFD shapes and suction-peak locations at every angle. For the Ahmed body, the measured drag coefficient is within 9% of the with-stand CFD, and the pressure distribution reproduces the nose suction peak, roof plateau, roof–slant suction peak and slant recovery. The main departures are a lift deficit at 5° and 10° , drag errors that change sign with angle of attack, and weaker measured suction on both models. These are discussed below.

4.2 Drag and Stand Interference

Comparing the two CFD cases isolates the drag added by the stand (sting and mount), including its interference with the model. Table 4.1 lists this increment together with the measured drag in excess of the stand-free CFD.

Table 4.1: Drag added by the stand. CFD increment = $C_d^{\text{with stand}} - C_d^{\text{without stand}}$, also given as a force at the test dynamic pressure and as a share of the with-stand C_d . WT excess = $C_d^{\text{WT}} - C_d^{\text{without stand}}$.

Case	CFD stand increment			WT excess ΔC_d	WT – CFD (with stand)
	ΔC_d	Force (N)	Share of C_d		
NACA, -5°	0.033	0.078	46%	0.025	–0.009
NACA, 0°	0.032	0.074	43%	0.034	+0.002
NACA, 5°	0.020	0.046	23%	0.026	+0.006
NACA, 10°	0.014	0.033	10%	0.027	+0.013
Ahmed body	0.173	0.040	28%	0.228	+0.055

The stand is a large part of the measured drag: 43–46% of the with-stand CFD drag at -5° and 0° , and 28% for the Ahmed body. This is why the stand-free CFD under-predicts the measured drag by 22–80% on the NACA 4418 and by 52% on the Ahmed body, while the with-stand CFD agrees to within the WT uncertainty in every case. Modelling the physical support is therefore essential when CFD is used to validate a small tunnel with a sting-mounted model.

The stand's contribution is not constant in the CFD. Its increment is steady at about 0.032 (0.075 N) for -5° and 0° but halves by 10° (0.033 N). The sting sits below the airfoil, in the flow shaped by its lower surface: at negative incidence that surface accelerates the flow (lower-surface C_p down to -0.9 at -5°), while at 10° it decelerates it ($C_p > 0$), so the sting sees a higher local dynamic pressure at negative angles and a lower one at positive angles. The measured excess over the stand-free CFD, by contrast, is nearly constant at 0.028 ± 0.004 (mean $\pm$ SD), a scatter well inside the WT uncertainty. The result is the WT–CFD error trend of Table 3.2: -12% at -5° rising steadily to $+9\%$ at 10° . (If a constant stand contribution of 0.028 is instead subtracted from the WT data, the remainder agrees with the stand-free CFD to within -8% to $+14\%$; the present data cannot distinguish the two interpretations, and a direct measurement of the stand alone would settle it.)

A likely cause of this angle-dependent error is the location of the angle-of-attack pivot, about 38 mm below the chord line rather than near the model's centre of gravity. Changing the angle therefore also moves the model forward or aft, and the balance was tared at the set angle before the fan was started. Once the flow is on, the aerodynamic loads act well above the pivot and produce a moment about it that the linkage can pass on to the drag lever, and the model was observed to shift slightly under load, moving

its weight away from the tared position. At negative incidence the model leans forward and pulls on the drag lever, lowering the reading; at positive incidence the lift and drag push it into the drag load cell, raising the reading. Because the NACA 4418 is cambered and the loads grow with angle, this bias is neither symmetric about 0° nor constant, consistent with the measured trend. Calibration adds a second, related error. Each load cell was calibrated once, with known weights hung vertically or horizontally on the empty lever system, while the balance was re-tared at the start of every angle. The tare only resets the zero (the intercept of the calibration line); it does not correct its slope. That slope is valid only if the load reaches the lever system through the same geometry as the calibration weights. Once the model is mounted, moved forward or aft by the angle change and displaced further under load, the point at which its loads and weight act, and hence the effective lever arm, differs from the calibration. The resulting error is a gain error: it scales with the load itself, so it cannot be removed by taring and grows with angle of attack. For the same reason, the one-axis-at-a-time calibration never measured how much of the lift appears on the drag channel and vice versa. A combined-load calibration with the model mounted at each test angle would quantify both effects.

4.3 Lift Deficit at Higher Angles of Attack

The measured lift falls 15% and 20% below the with-stand CFD at 5° and 10° , 2.6 and 4.0 times the WT uncertainty. Expressed as a lift-curve slope, the WT slope is 0.75 of the CFD slope in both angle intervals (0.036 vs. 0.048 and 0.052 vs. $0.069/^\circ$). A deficit that scales with angle in this way points to effects that grow with incidence or load rather than to a fixed offset. Three contributions are likely, and the data cannot separate them.

Low-Reynolds-number boundary-layer behaviour. At $Re \approx 1.2 \times 10^5$ the boundary layer on a thick cambered section is largely laminar over the forward chord and prone to laminar separation and trailing-edge separation, which reduce the lift-curve slope. The $k-\omega$ SST model used here treats the boundary layer as fully turbulent from the leading edge [9], so it keeps the flow attached and over-predicts lift. The weaker measured suction near the trailing edge (the $x/c = 0.75$ tap reads 0.16–0.23 less suction at every angle) is consistent with a thicker or partly separated boundary layer there. Transition modelling was outside the scope of this project.

Effective angle of attack. After the servo-driven angle mechanism failed, the angle was set by hand against a protractor. At the measured slopes, the lift deficit corresponds to an angle error of about 1° at 5° and about 2° at 10° . A setting error of up to a degree is plausible, and the loads, which grow with angle, can also deflect the model slightly away from the set angle once the fan is on (Section 4.2). The direction of that deflection was not measured, but a nose-down rotation would reduce lift in proportion to load, as observed, and would also explain why the 10° lower-surface pressures read slight suction where the CFD predicts positive C_p (Section 4.4).

Balance calibration. The lift channel is subject to the same gain error as the drag channel (Section 4.2): the calibration slope was obtained with weights on the empty lever, and the tare at each angle corrects only the zero. Because the forward–aft shift of the model grows with angle, the effective lever arm, and so the true gain, departs further from the calibrated value at higher incidence. An error of this kind is proportional to the load, which matches the uniform 0.75 ratio of the lift-curve slopes. Its size cannot be estimated without a calibration with the model mounted, but it is a plausible contributor alongside the two effects above.

Two further candidates can be ruled out as causes of the WT–CFD difference. Wall confinement and the gaps between the model’s ends and the side walls are present in both, because the CFD domain reproduces the confined test section with no-slip walls. Gravity, omitted in the CFD, only adds a hydrostatic pressure of about 0.5 Pa across the model height (0.3% of q_∞) in this incompressible flow, and the model’s weight is removed by the tare.

The lift deficit also explains the L/D behaviour: the measured L/D levels off between 5° and 10° (3.88 to 3.95) while the CFD value keeps rising, because the measured lift is reduced most at the angle where the measured drag is biased highest.

4.4 Surface Pressure Discrepancies

On both models the measured pressure distributions have the right shape but weaker suction than the CFD. On the Ahmed body the relationship is close to linear: a least-squares fit over the eight taps gives $C_p^{\text{WT}} \approx 0.65 C_p^{\text{CFD}} + 0.08$ ($R^2 = 0.95$, residual 0.04, comparable to the WT uncertainty). The measured values are therefore mostly a scaled-down copy of the CFD with a small positive shift. On the NACA 4418 the same fit gives a scale factor that varies with angle (0.59–0.93), and the lower surface departs from the pattern at 10° , so a single factor does not describe it. The candidate causes, and what the data say about each, are as follows.

Dynamic pressure. The C_p data were reduced with the nominal test speed, since the single differential sensor measured each tap against the Pitot static port and could not measure q_∞ at the same time. An error in q_∞ scales every C_p by the same factor. Within the ± 1.25 m/s speed uncertainty this gives up to about $\pm 14\%$, but a factor of 0.65 would need the true speed to be about 80% of the nominal value, roughly 13 m/s for the Ahmed test, which is well below the 16–19 m/s observed. A q_∞ error can contribute but cannot explain the full difference.

Reference pressure. The WT pressures are referenced to the Pitot static port, while the CFD C_p uses the solver's reference pressure. Any difference between these two references shifts every C_p by a constant. The fitted shifts are small (+0.08 for the Ahmed body, about zero for the pooled NACA data), so this effect is minor.

Pressure-line leakage. A leak between a tap and the sensor mixes the tap pressure with the pressure at the leak. If the leak opens into the test section, where the pressure is close to the static reference ($C_p \approx 0$), every reading is pulled towards zero by the same fraction, which produces exactly the scaling observed. The NACA 4418 routes each tap through internal 3D-printed channels; one of its lower taps ($x/c = 0.68$) was unusable, which shows that these channels are not always sealed. The Ahmed body used tubes brought directly to the taps, yet it shows a similar reduction, so the channels are unlikely to be the only path; tube joints and the tube passages through the test-section wall are other candidates. A leak test of each line would confirm or rule this out.

Real aerodynamic differences. On the NACA 4418 part of the C_p difference is physical rather than instrumental. A lower effective angle of attack and a thicker, partly separated low- Re boundary layer (Section 4.3) both weaken the upper-surface suction peak, which is where the largest differences occur (0.45–0.51 at the two leading upper taps at 10°). A lower effective angle also explains the sign difference on the lower surface at 10° , where the WT reads slight suction and the CFD a small positive pressure.

Other effects are smaller. The 4.9 Pa sensor resolution corresponds to $\Delta C_p \approx 0.03$. Air drawn in through unsealed joints and holes in the test section (the induced-draft fan keeps it below atmospheric pressure) may disturb the local flow near some taps.

4.5 Limitations

Flow quality. The empty test section reaches about 19 m/s and the speed fluctuates by roughly ± 1.25 m/s, but the spatial uniformity and turbulence intensity of the flow were not measured. Several features of the build could degrade them. Some straws in the honeycomb were squashed into elliptical cells during assembly. The only turbulence screen upstream of the model sits at the contraction exit rather than in the settling chamber, where screens are normally placed [4]: at that position the local dynamic pressure is $6.25^2 \approx 39$ times higher, so the screen's pressure loss is correspondingly larger and its wake reaches

the test section with little distance to decay. **[TODO: screen mesh size]** The joints between sections are sealed with epoxy fillets on the outside only, and air leaks in through them and through holes in the test section.

Wall effects. The model blocks about 10% of the test-section area (stand included). No blockage correction was applied [5]. Because the CFD reproduces the confined test section, the WT-CFD comparison is unaffected, but the coefficients reported here are confined-flow values and cannot be compared directly with free-air data.

Measurements. The fan speed was set with an open-loop regulator, without feedback control, and drifted between about 16 and 18.5 m/s. The C_p data were reduced with the nominal speed rather than a simultaneously measured one. All coefficients are normalised with the reference density $\rho_{\text{ref}} = 1.225 \text{ kg/m}^3$, the value used in the CFD, whereas the tunnel air at about 30 °C has $\rho \approx 1.164 \text{ kg/m}^3$. At a given measured speed this overstates q_∞ by about 5% and lowers every WT coefficient by the same fraction. The bias is small compared with the $\pm 14\%$ uncertainty in q_∞ for the pressure data, but it accounts for a few percentage points of the lift and suction deficits and slightly lowers the measured drag. The angle of attack was set manually, and small changes (below 1°) noticeably affected the readings. Pressure taps were switched by hand. The force balance was calibrated one axis at a time on the empty levers and re-tared at each angle, which corrects the zero but not the gain; its sensitivity to the model's position and its cross-coupling are unknown.

3D-printed models and balance. Both models were printed in PLA on a Bambu Lab A1 FDM printer. The printed surfaces carry layer lines and small dimensional errors in the profile, whereas the CFD geometry is exact and hydraulically smooth. At $Re \approx 1.2 \times 10^5$ the boundary layer is sensitive to roughness, which can move transition and separation and so change both the pressure distribution and the forces (Section 4.3). Printed tap holes may also be slightly off-normal or have rough edges, which biases the static pressure they sample. The lever arms of the dual-lever balance (rectangular cross-section) and the shafts connecting them to the bearings are also printed in PLA. They can flex slightly under load, and PLA creeps under sustained load, so the lever geometry, and with it the calibration gain, may differ between calibration and testing and drift during a run. Play between the printed shafts and the bearings adds further hysteresis. Stiffer materials (machined aluminium or steel levers and shafts) would remove most of these effects.

CFD. All simulations are steady RANS with $k-\omega$ SST. Several cases, especially those with the stand, did not converge in the residuals because the flow is mildly unsteady. They were run until the C_l , C_d and key C_p monitors were flat, or, where the monitors oscillated, averaged over whole cycles, with the spread reported as 1 SD. The student licence's cell limit restricted mesh refinement (15–18 boundary layers). The 10° with-stand case used additional wake refinement, so part of its difference from the stand-free case may be due to the mesh rather than the stand. No mesh-independence study, transient simulation or turbulence-model comparison was carried out.

4.6 Challenges Faced and Solutions

Table 4.2 summarises the main problems met during design, construction, testing and simulation, and how each was resolved.

Table 4.2: Challenges faced and how they were resolved.

Challenge	Problem	Solution
Fan and airflow	Four 185 CFM server fans in parallel, with closed-loop PID speed control, were planned for at least 10 m/s. They gave only 4–5 m/s: rated airflow applies at zero pressure rise, and the honeycomb, screens and contraction add losses.	Replaced by a 14 in, 350 W induced-draft drum fan (2000+ CFM) and the feedback control was dropped. The tunnel now reaches 16–18 m/s with a model installed.
Contraction cone	The fifth-order polynomial profile [6] cannot be formed from plywood, and the cone is too large for a standard 3D printer; several large prints failed.	Printed in eight flanged segments and joined with screws.
Flow straightener	A 3D-printed honeycomb would have thick walls and add blockage.	Built from thousands of drinking straws cut to length with a hot wire ($L/D \approx 7$).
Sealing	Air leaked through the joints between sections.	Epoxy fillets along the outer seams (not between flanges); some leakage remains (Section 4.5).
Angle of attack	The cam of the servo-driven angle mechanism broke.	Angle set manually with a protractor.
Tap selection	A cam-driven pinch-off selector, meant to open one tube at a time, jammed because of friction between the tubes and the 3D-printed cam.	Taps switched by hand, one tube at a time.
Pressure channels	Internal 3D-printed channels in the models were unreliable; one NACA lower tap failed.	The Ahmed body was built with tubes entering directly at the taps.
Balance friction	Stiction at the lever pivots prevented readings from returning to zero.	Lever pivots on 624ZZ ball bearings.
Pressure zero drift	The sensor's zero offset drifted between runs.	Automatic tare at the start of every run (fan off, mean of about 100 samples).
Flow visualization	A 24 V ultrasonic mist generator produced too little mist to be visible at test speed.	Smoke from glycerine-soaked cotton heated with a soldering iron.
CFD reference case	Stand-free CFD, representing the ideal model alone, left large gaps in the drag comparison.	The stand was added to the CFD geometry, so the with-stand runs act as a digital twin of the actual setup and test the measurement system itself.
CFD post-processing	Force reports picked up tunnel-floor faces; C_p from manually placed points was wrong at some taps.	Each wall zone was checked individually; C_p was read from the tap-plane curve.
CFD convergence and mesh	Steady solutions oscillated instead of converging, and the student cell limit restricted refinement.	Monitors run until flat, or averaged over whole cycles; local sizing at thin trailing edges and junctions, 15–18 boundary layers.

5 Conclusion and Future Work

5.1 Conclusions

A miniature open-circuit subsonic wind tunnel was designed, built and instrumented from laser-cut plywood, 3D-printed parts and low-cost electronics. With a model installed it runs at 16–18 m/s in a 160 mm × 160 mm test section, giving $Re \approx 1.2 \times 10^5$ on a 100 mm chord, and it measures lift, drag, surface pressure and freestream speed, with smoke visualization for qualitative flow observation. The measurements were compared with a CFD digital twin of the confined test section that includes the model's support stand.

The tunnel captures the aerodynamic trends of both test models. For the NACA 4418, the measured zero-lift angle (-3.7°) is within 0.1° of the CFD value, the lift at 0° agrees to within 1%, both data sets show the lift-curve slope rising between 5° and 10° , and the measured drag lies within -12% to $+9\%$ of the with-stand CFD. The measured pressure distributions reproduce the computed shapes and suction-peak positions on both models, and the Ahmed-body drag coefficient is within 9% of the with-stand CFD.

The magnitudes differ in three consistent ways, each with an identified cause. The measured lift is 15–20% below the CFD at 5° and 10° , attributed to low-Reynolds-number boundary-layer behaviour that the fully turbulent RANS model does not capture, to the effective angle of attack under load, and to the calibration of the balance. The drag error changes sign with angle of attack, consistent with coupling through the balance pivot. The measured suction is weaker than the CFD on both models, consistent mainly with attenuation in the pressure lines. None of these appears to be random error, so each can be reduced by the improvements below.

The main lesson for small facilities is that the support hardware must be part of the comparison. The stand accounts for 43–46% of the airfoil's drag at low incidence and 28% of the Ahmed body's; omitting it from the CFD makes the measured drag appear 22–80% too high, while including it brings the two into agreement within the measurement uncertainty. In a small, confined tunnel, modelling the actual stand and walls matters as much as the choice of turbulence model.

5.2 Future Recommendations

In order of expected benefit:

1. **Balance calibration and stiffness.** Calibrate with the model mounted at each test angle and with combined lift and drag loads, to measure the gain and cross-coupling errors directly. Replace the PLA levers and shafts with machined aluminium or steel parts, and move the angle-of-attack pivot close to the model.
2. **Stand-drag measurement.** Measure the drag of the stand alone, so that the stand's contribution can be removed from the balance readings instead of being modelled.
3. **Pressure system.** Leak-test every pressure line, avoid printed internal channels in future models, add a second differential sensor (or a pressure scanner) so that the dynamic pressure is measured together with each tap, and move to a 16-bit ADC (for example the ADS1115) for finer pressure resolution.
4. **Angle of attack.** Replace the protractor with a locking, indexed angle mechanism or a digital inclinometer, and check the angle with the flow on.
5. **Speed control.** Close the loop between the Pitot reading and the fan regulator, so that every run is held at the same dynamic pressure.
6. **Flow quality.** Survey the empty test section with a Pitot traverse and a hot-wire probe to measure uniformity and turbulence intensity; move the contraction-exit screen into the settling chamber, and seal the flange joints with gaskets.
7. **Wall corrections.** Apply blockage corrections [5] so that the results can also be compared with free-air data.
8. **CFD.** Carry out a mesh-independence study, use a transition-sensitive turbulence model and transient simulations for the oscillating cases, and extend the domain upstream to include the screens and

contraction.

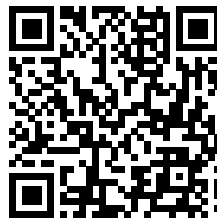
9. **Flow visualization.** Use smaller, smoother smoke nozzles with passive (suction-driven) injection, and record all test angles.
10. **Further models.** Reintroduce the circular cylinder, dropped during the project, as a benchmark with a well-known pressure distribution, and test the Ahmed body at other slant angles.

Data and Code Availability

All files needed to rebuild the tunnel and reproduce the results are openly available in the project's GitHub repository (link and QR code below). The repository contains:

- SolidWorks 2026 models and assemblies of the tunnel, force balance, test models, Pitot stand and smoke rake, with DXF files for laser cutting the plywood and acrylic panels;
- the Arduino firmware for the force balance and the surface-pressure survey;
- the MATLAB data-reduction and plotting scripts, together with the experimental coefficient workbook;
- the Ansys Fluent 2026 R1 project and meshing-workflow files, the simulated geometry (STEP), the force and pressure monitor histories, and the tap-plane pressure exports.

The Fluent case and data files (about 12 GB) are not included; they can be regenerated from the meshing-workflow files and the settings in Table 2.3. The repository README lists the folder layout and the software versions required.



<https://github.com/0xSYNDEED/PROJECT-WIND-TUNNEL>

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A Key Code Excerpts

The listings below are short excerpts of the data-acquisition firmware (`LiftDrag_V2.ino`, C++ for the Arduino Mega) and of the MATLAB data-reduction scripts, chosen because the methods in [Section 2](#) depend on them. Some lines are condensed and unrelated lines are replaced by `...`; the complete files are in the project repository (see [Data and Code Availability](#)).

Load-cell calibration and tare. Each channel is calibrated by a least-squares line through the recorded (raw reading, mass) points, and only the slope is stored. The tare at the start of each test re-sets the intercept so that the current reading becomes zero; it does not change the slope, which is the limitation discussed in [Section 4.2](#).

Listing A.1: Least-squares calibration slope and tare intercept (`LiftDrag_V2.ino`, excerpt; full code in the repository).

```

1 // runCalibration(): straight-line fit of mass [g] against raw HX711 counts
2 float denom = pointCount * sumX2 - sumX * sumX;
3 calculatedSlope = (pointCount * sumXY - sumX * sumY) / denom;
4 calculatedIntercept = (sumY - calculatedSlope * sumX) / pointCount;
5 ...
6 // doLoadCellTare(): intercept = -slope * rawTare => mass = 0 at the tared reading
7 interceptLift = -slopeLift * rawTareLift;
8 interceptDrag = -slopeDrag * rawTareDrag;
```

Tap pressures from the CFD tap-plane curve. Fluent exports the pressure coefficient along the tap plane as scattered points. The script finds the leading and trailing edges, rotates the points into the chord frame (which handles every angle of attack the same way), splits them into upper and lower surfaces, and interpolates at the tap positions.

Listing A.2: Chord-frame split of the tap-plane curve and interpolation at the taps (`NACA_Cp_Plots.m`, excerpt; full code in the repository).

```

1 x = T{:,2}; y = T{:,3}; cp = T{:,5}; % columns: node, x, y, z, Cp
2 [~, iTE] = max(x); % TE = most downstream point
3 d2 = (x - x(iTE)).^2 + (y - y(iTE)).^2;
4 [~, iLE] = max(d2); % LE = farthest point from TE
5 c = sqrt(d2(iLE));
6 ch = [x(iTE) - x(iLE), y(iTE) - y(iLE)] / c; % unit chord
7 nh = [-ch(2), ch(1)]; % unit normal, upper side positive
8 xc = ((x - x(iLE))*ch(1) + (y - y(iLE))*ch(2)) / c; % x/c along the chord
9 side = (x - x(iLE))*nh(1) + (y - y(iLE))*nh(2);
10 up = side >= 0; lo = ~up;
11 [xu, cpu] = cleanSort(xc(up), cp(up)); % sort, merge duplicate x
12 [xl, cpl] = cleanSort(xc(lo), cp(lo));
13 tapCpU = interp1(xu, cpu, tapU); % CFD Cp at the tap positions
14 tapCpL = interp1(xl, cpl, tapL);
```

Averaging a limit cycle. For the stand-free 10° case ([Fig. 2.8](#)), each cycle of the drag coefficient has two unequal peaks, so cycles are marked by the taller peak. All monitors are averaged over the last four whole cycles, so that the window starts and ends at the same phase, and the mean is accepted only if it changes by less than 0.5% between the last two and the last four cycles.

Listing A.3: Whole-cycle averaging of a limit-cycle solution (`AoA10_OscillationAverage.m`, excerpt; full code in the repository).

```

1 cd = Y(:,1); % drag-coefficient monitor
2 isPk = islocalmax(cd, 'MinProminence', 0.002, 'MinSeparation', 30) & it > skip;
3 pk = find(isPk);
4 tall = pk(cd(pk) > mean(cd(pk))); % one marker per cycle
5 n = min(nUse, numel(tall) - 1);
6 w = tall(end-n) : tall(end)-1; % whole cycles, same phase at both
7 ends
8 mu = mean(Y(w,:)); sd = std(Y(w,:)); % reported mean and spread
9 nh = max(1, floor(n/2)); wh = tall(end-nh) : tall(end)-1;
```

```
9 shiftPct = abs(mean(Y(wh,1:2)) - mu(1:2)) ./ abs(mu(1:2)) * 100; % must be < 0.5 %
```

Averaging an irregular oscillation. The Ahmed body's slant pressures oscillated without a clear period, so cycles cannot be counted. The monitors are instead averaged over a fixed window after the start-up transient, and the window is accepted if the means of its two halves agree.

Listing A.4: Fixed-window averaging with a settling check (Ahmed_CFD_Oscillation_Handling.m, excerpt; full code in the repository).

```
1 w = it >= i1 & it <= i2; % window after the transient
2 iw = find(w);
3 Yw = Y(w,:);
4 mu = mean(Yw)'; sd = std(Yw)'; % mean and spread of Cd and the 8 tap Cp
5 h = floor(numel(iw)/2); % has the mean settled? compare the two halves
6 m1 = mean(Y(iw(1:h),:)); m2 = mean(Y(iw(h+1:end),:));
7 fprintf('Cd mean shift between halves: %.2f%%\n', abs(m2(1)-m1(1))/abs(m1(1))*100);
8 fprintf('Max Cp mean shift between halves: %.4f\n', max(abs(m2(2:end)-m1(2:end))));
```

Uncertainty propagation. The wind-tunnel error bars follow Eq. (4). For L/D the dynamic pressure cancels, so only the force terms remain.

Listing A.5: Uncertainty of the force and pressure coefficients (NACA_Force_Plots.m and NACA_Cp_Plots.m, excerpts; full code in the repository).

```
1 % force coefficients (NACA_Force_Plots.m)
2 q = 0.5*rho*U^2; dq_q = 2*dU/U;
3 dL = dL_g/1000*gAcc; dD = dD_g/1000*gAcc; % force resolution [N]
4 L = M.C1*q*A; D = M.Cd*q*A;
5 dC1 = sqrt((dL/(q*A))^2 + (M.C1*dq_q).^2);
6 dCd = sqrt((dD/(q*A))^2 + (M.Cd*dq_q).^2);
7 dLD = abs(LD(:,1)) .* sqrt((dL./L).^2 + (dD./D).^2);
8 % pressure coefficients (NACA_Cp_Plots.m)
9 dCpFun = @(cp) sqrt((dp/q)^2 + (cp^2*dU/U).^2);
```

B NACA 4418 Surface Pressure Data

Measured and computed (CFD with stand) pressure coefficients at each tap, with the WT uncertainty of Section 2.5. The faulty lower tap at $x/c = 0.68$ is omitted.

Table B.1: NACA 4418 tap pressure coefficients: wind tunnel vs. CFD (with stand).

α	Surface	x/c	C_p^{WT}	$\pm$	C_p^{CFD}	WT-CFD
-5°	Upper	0.05	0.251	0.044	0.306	-0.056
	Upper	0.20	-0.293	0.049	-0.470	+0.176
	Upper	0.45	-0.384	0.061	-0.539	+0.155
	Upper	0.75	-0.091	0.029	-0.249	+0.158
	Lower	0.12	-0.666	0.099	-0.915	+0.249
	Lower	0.27	-0.352	0.057	-0.549	+0.197
	Lower	0.52	-0.176	0.036	-0.272	+0.096
0°	Upper	0.05	-0.219	0.041	-0.184	-0.035
	Upper	0.20	-0.565	0.085	-0.796	+0.231
	Upper	0.45	-0.453	0.070	-0.669	+0.216
	Upper	0.75	-0.064	0.028	-0.245	+0.181
	Lower	0.12	-0.298	0.050	-0.602	+0.303
	Lower	0.27	-0.203	0.039	-0.421	+0.219
	Lower	0.52	-0.128	0.032	-0.232	+0.104
5°	Upper	0.05	-0.720	0.106	-0.723	+0.003
	Upper	0.20	-0.800	0.117	-0.988	+0.188
	Upper	0.45	-0.560	0.084	-0.677	+0.117
	Upper	0.75	-0.043	0.027	-0.202	+0.159
	Lower	0.12	-0.107	0.030	-0.190	+0.083
	Lower	0.27	-0.133	0.032	-0.194	+0.061
	Lower	0.52	-0.101	0.030	-0.117	+0.015
10°	Upper	0.05	-0.933	0.136	-1.443	+0.510
	Upper	0.20	-0.880	0.128	-1.331	+0.451
	Upper	0.45	-0.554	0.083	-0.770	+0.216
	Upper	0.75	0.032	0.027	-0.201	+0.233
	Lower	0.12	-0.048	0.027	0.171	-0.219
	Lower	0.27	-0.112	0.031	0.034	-0.146
	Lower	0.52	-0.101	0.030	0.013	-0.114